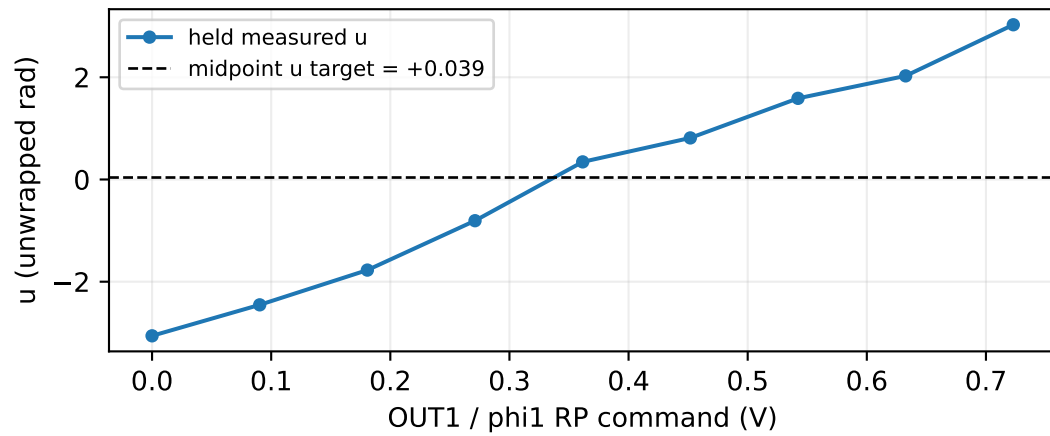
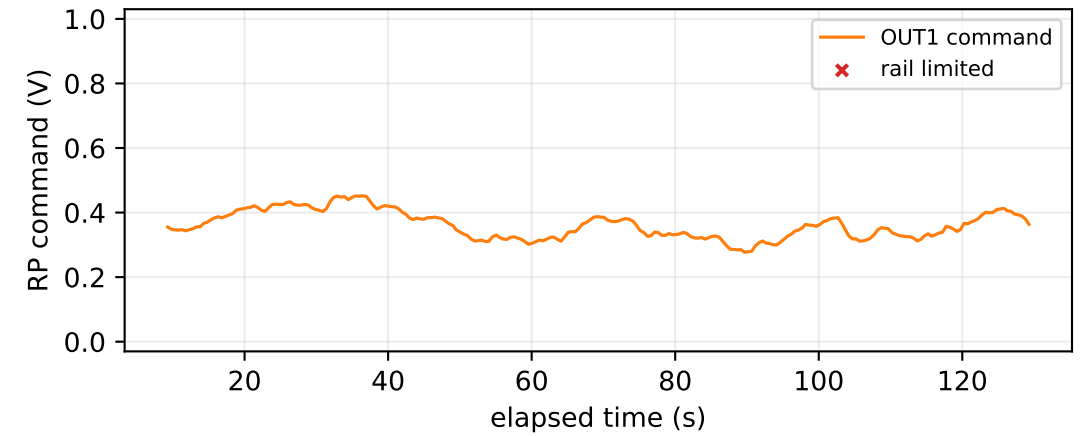


Phi1 u-lock at first-NPBS D — 234652_phi1-lock-test_phi1-lock-1

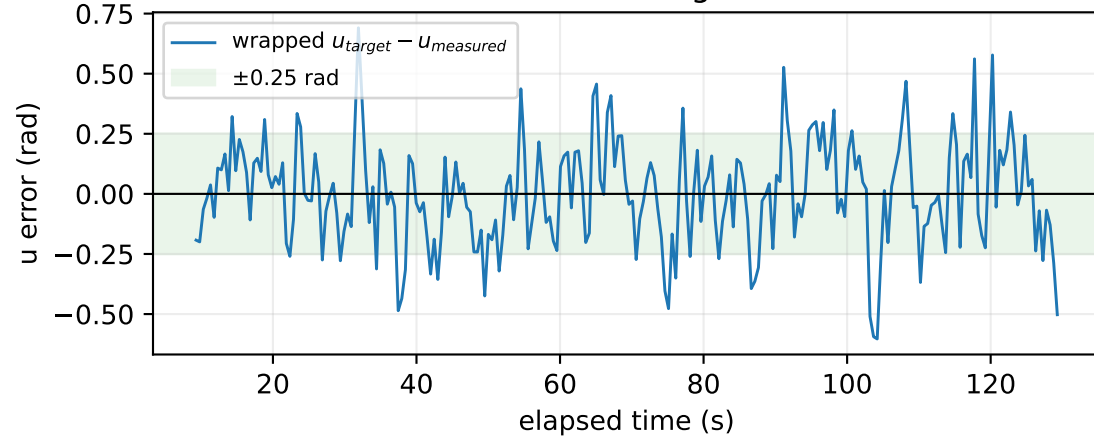
Fresh one- $V\lambda$ u calibration



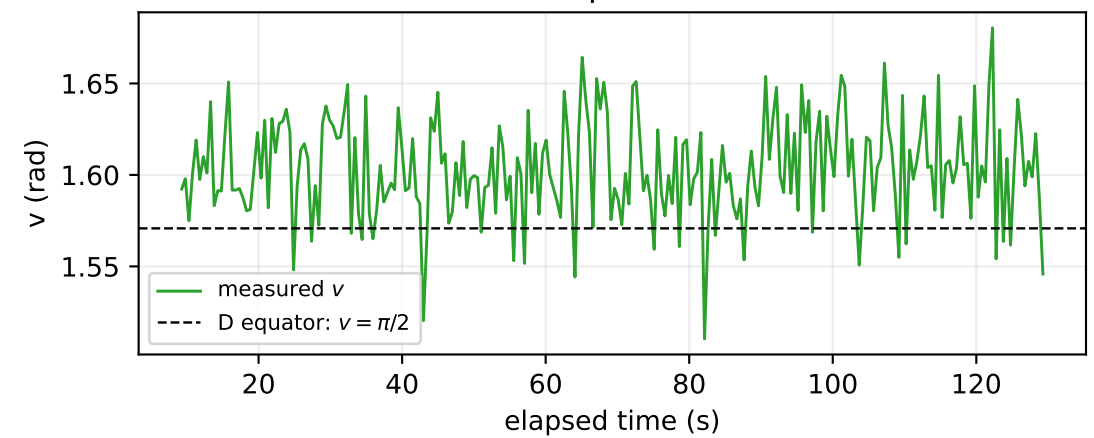
Phi1 actuator command



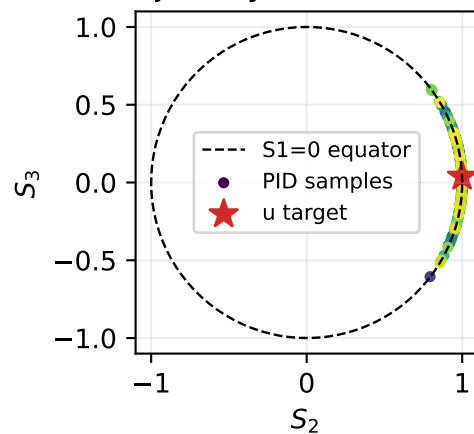
Azimuth locking error



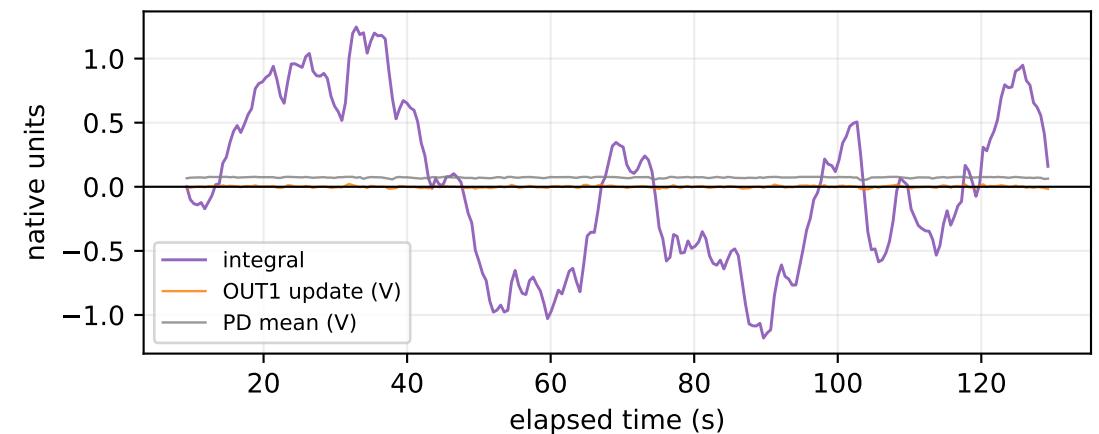
Uncontrolled polar coordinate



Held lock trajectory on the Poincare equator



PI state, command increment, and PD



Target u: +0.0387 rad Calibrated du/dV : +8.506 rad/RP V $V\lambda$ estimate: 0.7386 RP V $K_p=0.280$ $K_i=0.0050/s$ Stokes average=3
 Duration: 120.0 s Median |error|: 0.135 rad (full), 0.169 rad (late) v excursion (5-95%): 0.088 rad mean $|S_1|$: 0.036 Rail-limited: 0/240